Chapter 1

Introduction

Commutative algebra is about studying commutative rings and their modules. The course has a view to algebraic geometry.

Example 1.1 (Examples of Commutative Rings). $\mathbb{Z}$, $k[x_1, ..., x_n]$ for k field. For topological spaces, X we get the commutative ring $C(X) := \{f : X \to \mathbb{C} : f \text{ continous}\}$.

Theorem 1.1. Structure Theorem for Finitely Generated Rings over a PID

Theorem 1.2. $k[x_1, ..., x_n]$ is a UFD

Chapter 2

Basic Definitions

Definition 2.1 (Ring). A **ring** A is a triple $(A, +, \times)$ where A is a set and $+$ and $\times$ are binary operations on A.
1. $(A, +, 0)$ is a abelian group
2. $(A, 1, \times)$ is a monoid.
3. Distributive Laws: $(a + b) \times c = a \times c + b \times c$ and $a \times (b + c) = a \times b + a \times c$

A is a **commutative ring** if and only if $(A, 1, \times)$ is a commutative monoid.

Example 2.1. Examples of noncommutative rings are $M_n(k)$ where k is a field, kG for a nonabelian group G. Examples of commutative rings include $C^\infty(X)$ for X smooth manifold.

If a ring is discussed, then it is commutative from now on unless otherwise stated.

Note on the definition, $1 \in A$ (by definition), though this is not not always required. A zero ring: $0 = 1 \in A$

Exercise 2.1. Show $\forall x \in A$, $0 \cdot x = 0$ and $(-1)x = -x$

Definition 2.2 (Unit). A element $x \in A$ is called a unit (or invertible) if $\exists y \in A$ where $xy = 1 = yx$. An exercise is showing that y is unique. $A^\times$ is the group of units of A.

Definition 2.3 (Zero Divisor). A element $x \in A$ is a **zero divisor** if there exists a nonzero $y \in A$ where $xy = 0$.

Definition 2.4 (Nilpotent). $x \in A$ is **nilpotent** if $\exists n \in \mathbb{N}$ where $x^n = 0 \in A$

Definition 2.5 (Field). A is a field if $1 \neq 0$ and for every $x \neq 0$ in A is a unit.

Definition 2.6 (Integral Domain). A is a **integral domain** or **domain** if $1 \neq 0$ and $\forall x, y \in A$ $xy = 0 \Rightarrow x = 0$ or $y = 0$.

Definition 2.7 (Reduced). A ring A is **reduced** if only nilpotent element in A is 0.

We have the following inclusions:

$$\text{Field} \subset \text{Domain} \subset \text{Reduced}$$

Example 2.2. The zero ring is reduced, but not a domain (or a field). $\mathbb{Z}_p$ is a field if and only if p is prime in $\mathbb{Z}$. $\mathbb{Z}_n$ is not a field if and only if n is not prime. If $\mathbb{Z}_{p_1 \cdots p_n}$ is reduced if $n \geq 0$ and $p_1, ..., p_n$ are distinct.

Definition 2.8 (Subring). A subring of a ring R, $\{0, 1\} \subset A \subset R$ is a subset where A is closed under addition, additive inverses, multiplication, (in R)

Remark 2.1. $\{0\}$ is not a subring of $\mathbb{Z}$

Note in the category of commutative rings, $\mathbb{Z}$ is an initial object.

Definition 2.9 (Left and Right Ideal). An right ideal I in a ring A is a subset where $\forall k \in I$, then $kA \subseteq I$. A left ideal I is subset where $Ak \subset A$. In a commutative ring left and right ideals are equal. Kernel of a ring homomorphism is always a ring, furthermore any ideal is a kernel of a homomorphism

Example 2.3. In a field k, the only ideals are $\{0\}$ and k. If X is a topological space, $Y \subset X$ is a closed (if it isn't closed you can just take closure because of continuity) subspace. Let $I = \ker\left(C(X) \xrightarrow{|_Y} C(Y)\right)$ is a ideal. This is a typical way to get ideals, take a space of functions which is a ring and then restrict on a closed subspace.

Definition 2.10 (The Ideal Generated by a Set). The **ideal generated by $S \subset A$ a ring** denoted (S):
1. The intersection of all ideals containing S.
2. The set of finite linear combinations of elements of S. If $S = \emptyset$. Thus set is the 0 ideal.

For a finite set $S = \{a_1, ..., a_r\} \subset A$ we will used the notation: $(a_1, ..., a_r) = (S)$.

Definition 2.11 (Principal Ideal). An ideal generated by one element.

Note, while the intersection of two ideals is always a ideal, the union is not always a ideal. But the sum of two ideals is a ideal, in fact the smallest ideal containing $I \cup J$.

Definition 2.12 (Quotient Ring). Standard way: R/I is the set of cosets but not ideal Better way: A/I is a ring that come with a ring homomorphism $\varphi : A \twoheadrightarrow A/I$ with kernel I.

Note for the sake of notation: For $a \in A$ we can say $a \in A/I$ to talk about $a + I \in A/I$.

Definition 2.13 (Maximal Ideals). A ideal I is called maximal if A/I is a field.

Definition 2.14 (Prime Ideals). A ideal I is a prime if A/I is a domain

Definition 2.15 (Radical Ideal). A ideal I is radical if A/I is reduced.

Example 2.4 (Maximal, Prime, and Radical Ideas). For a field k, the maximal ideal in $k[x]$ (which is also a PID) is (f) where f an monic irreducible polynomial.

Monic means leading coeefficent is 1. In any **domain** R, $f \in R^\times$ and for $g, h \in R$ $f = gh \Rightarrow$ g or h is a unit for R.

Prime ideals of $k[x]$ are the maximal ideal and 0. Finally the radical ideals of $k[x]$ are (0) and $(f_1, ..., f_r)$ for $1 \leq r < \infty$

Example 2.5 (Prime ideals in a UFD). $\mathbb{Z}[x]$ is a unique factorization domain and not a pid. Consider $(0), (7), (x), (7, x) \subset \mathbb{Z}[x]$. Only $(7, x)$ is maximal.

Example 2.6. Let S be a set, k a field, and A a ring of functiosns $S \to k$ that contains all constant function. Fix $s_0 \in s$

In "good cases", in different points in S give different maximal ideals

Example 2.7. If X a compact Hausdorff space, or a metrizable topological space, by the Tietze Extension Theoerm for any closed subset $Y \subset X$ for any continous $f : Y \to \mathbb{R}$ there is a Extension $X \to \mathbb{R}$. Basically you can seperate points hence the ideals generated are different.

Chapter 3

Spectrum of Ring

Definition 3.1 (Spectrum of a ring). For a commutative ring A,

$$\mathrm{Spec}(A) := \{\text{prime ideals in A}\}$$

Think of elements of A as functions on the space $\mathrm{Spec}(A)$. So A turns into the ring of functions on $\mathrm{Spec}(A)$. For each $\mathfrak{p} \in \mathrm{Spec}(A)$, there is a homomorphism

$$A \to A/\mathfrak{p} \to \mathrm{Frac}(A/\mathfrak{p}).$$

For a function $f \in A$, the value of f at $\mathfrak{p}$ is the image of f in this homomorphism.

Why are we taking the fraction field of a $A/\mathfrak{p}$?

3.1 Why only prime ideals?

We want to take values in a field k. $A \overset{\text{hom}}{\to} k$ maps A to a subring of k, which is a domain, hence the map has a prime ideal as a kernel.

Example 3.1 (Spectrum of Basic Rings).
1. $\mathrm{Spec}\, 0 = \emptyset$
2. If $A \neq 0$, then $\mathrm{Spec}\, A \neq \emptyset$ since A must have a maximal ideal hence a prime ideal.

3.2 Zariski Topology

We will define a topology on $\mathrm{Spec}(A)$.

Definition 3.2 (Zariski Topology of $\mathrm{Spec}(A)$). For a ideal $I \subseteq A$, denote

$$V(I) := \{p \in \mathrm{Spec}(A) : I \subseteq p\}.$$

A subset of $\mathrm{Spec}(A)$ is closed if it is equal to $V(I)$ for some $I \subseteq A$ ideal. A subset of $\mathrm{Spec}(A)$ is open if its the complement of $V(I)$ for some $I \subseteq A$ ideal.

Why does $V(I)$ consist of ideals containing I rather being contained in I? Interpreting elements of A as functions on

$$V(I) := \{\text{pts where all functions in I vanish}\}$$

3.2.1 Nonstandard Notion:

For elements $f_1, ..., f_r \in A$ write $\{f_1 = 0, ..., f_r = 0\} \subseteq \mathrm{Spec}(A)$ for $V((f_1, ..., f_r))$

Theorem 3.3. Show that the closed and open sets defined before form a Topology.

Example 3.2 (Some more Spec).

1. $\text{Spec}(\mathbb{Q}) = \{(0)\}$
2. $\text{Spec}(\mathbb{R}) = \{(0)\}$
3. $\text{Spec}(\mathbb{Z}) = \{(0)\} \cup \{(p) : p \text{ prime}\}$.

$$\{10 = 0\} = \{(2), (5)\}$$

3. $\text{Spec}[\mathbb{C}[x]]$: $\mathbb{C}[x]$ is a PID, and thus we know how the prime ideals look. The prime ideal looks (0) or (f) where $f \in \mathbb{C}[x]$ monic irreducible polynomial. Since $\mathbb{C}$ is algebraic closed, we know that f has degree greater than 1 it can be factored. Thus f must be degree 1 polynomial. $\mathbb{C}[x]/(0)$ is a domain but not a field. $\mathbb{C}[x]/(x-b) \cong \mathbb{C}$. $\text{Spec}(\mathbb{C}[x])$ looks like $\mathbb{C}$ with a extra point (0). The closed subsets of X are the subsets

$$V(I) := \{\mathfrak{p} \in \text{Spec}(\mathbb{C}[x]) : I \subseteq \mathfrak{p}\}.$$

Note that $\mathbb{C}[x]$ is a PID, thus any ideal is $(0), (1), ((x-b_1)^{n_1}...(x-b_n)^{n_r})$ where $b_i \in \mathbb{C}$ and $r \geq 1$.

$$V((0)) = \{0 = 0\} := \text{Spec}(C[[x]])$$
$$V((1)) = \{1 = 0\} = \emptyset$$
$$V((x-b_1)^{n_1}...(x-b_n)^{r_n}) = \{(x-b_1)^{n_1}...(x-b_n)^{n_r} = 0\} = \{(x-b_1), ..., (x-b_n)\}$$

Thus the closed subsets of $\text{Spec}(\mathbb{C})$ looks like finite sets of $\mathbb{C}$ portion of $\text{Spec}[\mathbb{C}]$, empty set, and the whole set. Thus as a topological space, $\text{Spec}(\mathbb{C})$ looks exactly like $\mathbb{C}$ plus a extra point. (0) is called a generic point because the closure of the point is the whole space.

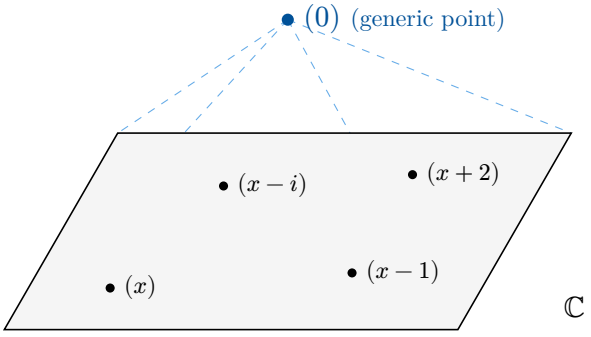


Figure 1: $\text{Spec}(\mathbb{C}[x])$: closed points $(x-b)$ for $b \in \mathbb{C}$, plus the generic point (0), whose closure is all of $\text{Spec}(\mathbb{C}[x])$.

Theorem 3.4 (Nilradical is intersection of all prime ideals). The nilradical, $\mathfrak{N}$, is the set of nilpotent elements of a ring.

$$\mathfrak{N} = \bigcap_{\mathfrak{p} \in \text{Spec}(A)} \mathfrak{p}$$

Proof:

1. $\mathfrak{R} \subseteq \bigcap_{\mathfrak{p} \in \operatorname{Spec}(A)} \mathfrak{p}$: This is trivial because $x \in \mathfrak{R} \Rightarrow x^n = 0$ and thus $x = 0 \in A/\mathfrak{p}$ because $A/\mathfrak{p}$ is a domain, hence $x \in \mathfrak{p}$.
2. $\mathfrak{R} \supseteq \cap_{\mathfrak{p} \in \operatorname{Spec}(A)} \mathfrak{p}$: We will go for a contrapositive. Suppose x is not nilpotent. Then you want $\mathfrak{p} \in \operatorname{Spec}(A)$ such that $x \notin \mathfrak{p}$. Consider the poset of $P = \{I \subseteq \operatorname{Spec}(A) : x^{\mathbb{N}} \notin I\}$ with the comparison relation given by $\subseteq$. Suppose $S = \emptyset$ empty chain, note that $P \neq \emptyset$ because $(0) \in P$. If S is a nonempty chain in P, let

$$J = \bigcup_{I \in S} I \subseteq A$$

is a ideal. Since no interim ideal contains any positive power of x, J doesn't contains any positive power of x. By Zorn's lemma there exists a maximal element I. Clearly $1 \notin I$, because $x \notin I$. Suppose $y \in R - I$, $z \in R - I$ and $yz \in I$. Thus $I \subset I + (y)$ and $I \subset I + (z)$. So $I + (z), I + (y)$

□

Definition 3.5 (Radical Ideal). A ideal I is **radical** if R/I has no nilpotent elements.

Definition 3.6 (Radical of a Ideal). For a ideal I in a ring R, the **radical** of I is

$$r(I) := \{x \in R : (\exists n \geq 1) x^n \in I\} = [R \to R/I]^{-1}(\mathfrak{R})$$

where $[R \to R/I]^{-1}(\mathfrak{R})$ is the inverse image of the nilradical of R/I. Equivalently $r(I)$, is the smallest radical ideal containing I.

Corollary 3.7. For any ideal $I \subseteq R$, $r(I)$ is the intersection of the prime ideals that contain I.

Theorem 3.8. For any ring R and ideals $I, J \subseteq R$. $V(I) = V(J) \Leftrightarrow r(I) = r(J)$.

Chapter 4

Module

Definition 4.1 (Left Module). A **Left Module** over a ring A (not necessarily commutative) is a abelian group M with a action of the ring A such that:
1. $1 * m = m$ for all $m \in M$
2. $(ab) * m = a(b * m)$
3. $(a + b) * m = a * m + b * m$

Remark 4.1. A right A-module M, written $m * a \in M$ for $m \in M$ and $a \in A$. They are not always equal to to the left A-module. For a commutative ring left and right modules are the same.

Example 4.1. For a field k, a k-vector space is a k-module. For a ring A and $n \geq 0$, A^n is a A-modules, called the free module. A $\mathbb{Z}$-module is equivalent to a abelian group. An ideal I in a commutative ring determines two obvious modules: I itself, A/I. In a more general setting, a left ideal is a left submodule of A, a right ideal is a right submodule. For k a field, a $k[x]$-module is a k-vector space M together with a k-linear map from $M \to M$ (action of x). More generally, a $k[x_1, ..., x_n]$ is a k vectorspace with n maps $x_1 : M \to M$, ... $x_n : M \to M$ such that the maps commute. If E is a real vector bundle over a topological space X, then the set M of continuous sections of E is a module over the $C(X)$.

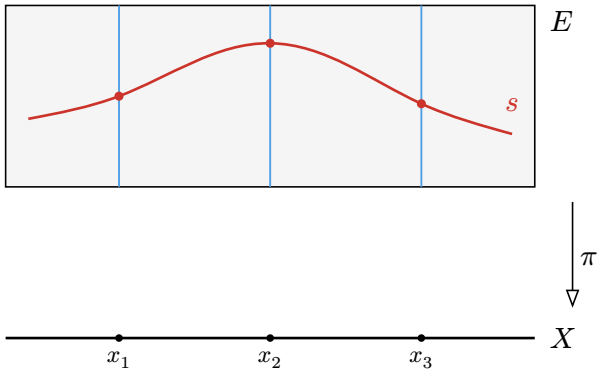


Figure 2: A real vector bundle $\pi : E \to X$. A section s picks one point $s(x)$ in each fiber E_x; the continuous sections form a $C(X)$-module.

When seeing the theorem about modules , consider what it tells you about ideals, by looking at A/I.

The category of A modules is a additive category, unlike (say) the category of commutative rings or commutative A-algebra. Thus A-modules form a nicer simpler category, and studying

A-modules is nicer than studying A. For a commutative ring, $\mathrm{Hom}_A(M, N)$ is a A-module (this is not true for a generic ring).

Definition 4.2. For a A linear map φ:

$$\ker(\varphi) := \{x \in M : \varphi(x) = 0\}$$
$$\mathrm{im}(\varphi) := \{f(m) : m \in M\}$$
$$\mathrm{coker}(f) = N/\mathrm{im}(\varphi)$$

Theorem 4.3 (Universal Property of Quotient Module). Given M a A-module and N a A-submodule, we get a quotient module M/N with a A linear map $\varphi : N \to M/N$ with $\ker \varphi = N$. Equivalently, for any module Q, there is a 1 to 1 correspondence between A-linear maps $N/M \to Q$ and A linear maps $N \to Q$ with kernel which contains M.

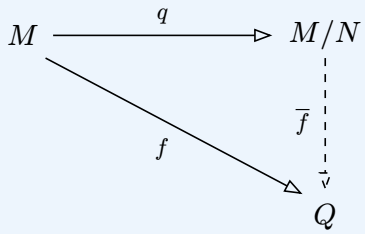


Figure 3: Universal property of M/N: any $f : M \to Q$ with $N \subseteq \ker f$ factors uniquely through the quotient.

Theorem 4.4 (Universal Property of Free Module). For a commutative ring A and any $n \geq 0$, there is a 1 to 1 correspondence from A-linear maps $A^n \to M$ with n-tuples of elements of M.

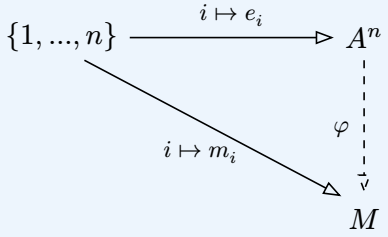


Figure 4: Universal property of A^n: any choice $e_i \mapsto m_i$ of n elements of M extends uniquely to an A-linear map φ.

For any ring homomorphism $A \xrightarrow{f} B$, we get a continuous map

$$\begin{aligned} \mathrm{Spec}\, B &\xrightarrow{f^*} \mathrm{Spec}\, A \\ \mathfrak{p} &\to f^{-1}(\mathfrak{p}). \end{aligned}$$

Thus Spec is a contravariant functor.

Theorem 4.5. For any ideal I in a commutative ring, the ring homomorphism $A \to A/I$ determines a continuous map

$$\operatorname{Spec}(A/I) \to \operatorname{Spec}(A)$$

is a homomorphism into a closed subset of $\operatorname{Spec}(A)$.

Proof: Follow your nose. □

Example 4.2 (What is $\operatorname{Spec}(Z/10)$?). By the theorem, the set is homomorphic to $\{10 = 0\} \subseteq \operatorname{Spec}(\mathbb{Z})$ in otherwords $\{(2), (10)\}$ with the discrete topology.

Lemma 4.6. Let R be a domain. Then $\overline{\{(0)\}} = \operatorname{Spec}(R)$.

Proof: Let S, be a closed subset which contains (0). $\exists I \subseteq R$ such that $S = V(I)$. So $I \subset (0) \Rightarrow I = (0)$. Then $S = V(I) = \operatorname{Spec} R$. □

Corollary 4.7. Let R be any commutative ring, and $\mathfrak{p} \in \operatorname{Spec}(R)$ a point. Then the closure of this point is $V(\mathfrak{p})$.

Proof: $\operatorname{Spec}(A/\mathfrak{p})$ is homomorphic to a closed subset $\operatorname{Spec}(A)$. $V(\mathfrak{p})$. But $(0) \in A/\mathfrak{p}$ comes from $\mathfrak{p}$. By 4.6 we can conclude the result, because $\overline{\{(\mathfrak{p})\}} \cong \overline{\{(0)\}} = \operatorname{Spec}(A/\mathfrak{p}) \cong V(I)$. □

Lemma 4.8. For any ring R, $\operatorname{Max} R \coloneqq \{\text{maximal ideals of } R\}$ is the set of closed points

Proof: Given a maximal ideal $\mathfrak{m} \in \operatorname{Spec} R$. $\overline{\{\mathfrak{m}\}} = V(\mathfrak{m}) = \{\mathfrak{p} \in \operatorname{Spec} R : \mathfrak{m} \subset \mathfrak{p}\} = \{\mathfrak{m}\}$. If $\mathfrak{p} \in \operatorname{Spec}(R)$ is closed, then $V(\mathfrak{p}) = \{\mathfrak{q} \in \operatorname{Spec}(R) : \mathfrak{p} \subseteq \mathfrak{q}\} = \{\mathfrak{p}\}$ hence $\mathfrak{p}$ is not contained in any other prime ideal, hence maximal ideal and must be maximal. □

Definition 4.9 (Irreducible Subspaces). A topological space X is irreducible if $X \neq \emptyset$ and you can't write $X = X_1 \cup X_2$ proper and X_1 and X_2 closed in X.

4.1 Operations on Modules.

Definition 4.10. For a subset S for a R-module M, the submodule generated by S is the intersection of all submodules containing.

Chapter 4.1

References